

Malware Analysis A5

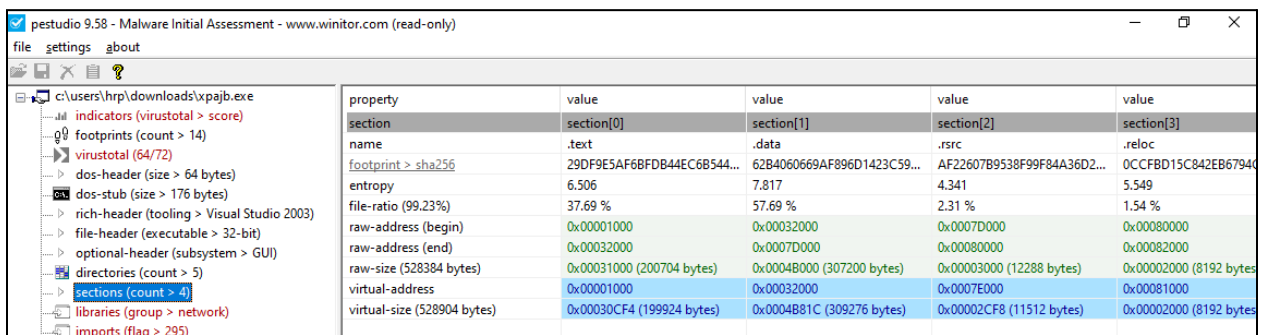
Objective:

Download a suspicious Portable Executable (PE) file from any previously mentioned source and perform a complete static analysis using and : determine if the file is packed through entropy analysis and section inspection, explain why packing information is critical for interpreting imports and exports, and identify at least two indicators of packing; list and analyze all sections (.text, .data, .rdata, etc.) with their size, permissions, and note any unusual sections; extract the compiler timestamp and discuss what it reveals about the file's origin; list all imported libraries, APIs, and functions (highlighting key ones such as kernel32.dll, user32.dll, ws2_32.dll) and map their functional significance (e.g., CreateRemoteThread → process injection, WinExec → external execution, SAsocketA → networking); check for any exported functions and infer their possible role in persistence or malware reuse; based on these findings, build a likely behavior profile of the malware (process injection, persistence, network communication, file system manipulation, or anti-analysis).

Additionally, compare the properties of any available files using the file and xxd commands, generate MD5 and SHA-256 hashes for both, and verify them with VirusTotal. Finally, pack the benign .exe, compare its structure, entropy, size, and imports before and after packing, then attempt to unpack it and repeat the analysis to demonstrate how packing alters the visibility of executable characteristics.

Downloaded Targeted malicious file - **xpajb.exe** from Da2Dalus/Github

Step 1: Static Analysis



The screenshot shows the PEStudio 9.58 interface for the file `c:\users\hrp\downloads\xpajb.exe`. The left sidebar lists various file properties, including indicators (VirusTotal score), footprints (count > 14), VirusTotal (64/72), DOS-header (size > 64 bytes), DOS-stub (size > 176 bytes), Rich-header (tooling > Visual Studio 2003), file-header (executable > 32-bit), optional-header (subsystem > GUI), directories (count > 5), sections (count > 4), libraries (group > network), and imports (flag > 295). The main pane displays a table of section properties.

property	value	value	value	value
section	section[0]	section[1]	section[2]	section[3]
name	.text	.data	.rsrc	.reloc
footprint > sha256	29DF9E5AF6BFD844EC6B544...	62B4060669AF896D1423C59...	AF22607B9538F99F84A36D2...	0CCFBD15C842EB67940
entropy	6.506	7.817	4.341	5.549
file-ratio (99.23%)	37.69 %	57.69 %	2.31 %	1.54 %
raw-address (begin)	0x00001000	0x00032000	0x0007D000	0x00080000
raw-address (end)	0x00032000	0x0007D000	0x00080000	0x00082000
raw-size (528384 bytes)	0x00031000 (200704 bytes)	0x00048000 (307200 bytes)	0x00003000 (12288 bytes)	0x00002000 (8192 bytes)
virtual-address	0x00001000	0x00032000	0x0007E000	0x00081000
virtual-size (528904 bytes)	0x00030CF4 (199924 bytes)	0x0004B81C (309276 bytes)	0x00002CF8 (11512 bytes)	0x00002000 (8192 bytes)

Filename: xpajb.exe

Tool used: PEStudio 9.58

VirusTotal Detection: 64/72 (highly malicious)

File Type: 32-bit executable (GUI subsystem)

We can see that the file is unpacked because we observe the .text, .data, etc. However, when examining PEStudio, we notice that the .data segment shows a higher entropy range (57%). This suggests it might be corrupted or encrypted. Typically, .data contains variables and has low entropy.

Step 2: Compiler Stamp

property	value	detail
characteristics	0x010E	
dynamic-link-library	0x0000	false
32-bit words support	0x0100	true
file-can-be-executed	0x0002	true
system-image	0x0000	false
large-address-aware	0x0000	false
debug-stripped	0x0000	false
line-stripped-from-file	0x0004	true
local-symbols-stripped-from-file	0x0008	true
relocation-stripped	0x0000	false
uniprocessor	0x0000	false
bytes-of-machine-words-reversed-Low	0x0000	false
bytes-of-machine-words-reversed-Hi	0x0000	false
media-run-from-swap	0x0000	false
network-run-from-swap	0x0000	false
general		
compiler-stamp	0x480381D3	Mon Apr 14 16:09:55 2008 UTC
size-of-optional-header	0x00E0	224 bytes
signature	0x00004550	PE00
machine	0x014C	Intel-386
sections-count	0x0004	4
pointer-symbol-table	0x00000000	0x00000000
number-of-symbols	0x00000000	0x00000000

- **Timestamp (from PE header):**
0x480381D3 → Mon Apr 14 16:09:55 2008 UTC

Interpretation:

- Timestamps can be forged by packers.
- The 2008 date may be legitimate or fake. If VirusTotal detects it as modern malware, a 2008 timestamp suggests tampering.
- Used in threat intelligence: Analysts often compare timestamps with known malware campaigns.

pestudio 9.58 - Malware Initial Assessment - www.winitor.com (read-only)

file settings about

c:\users\hrp\downloads\xpajb.exe

indicators (virustotal > score)

footprints (count > 14)

virustotal (64/72)

dos-header (size > 64 bytes)

dos-stub (size > 176 bytes)

rich-header (tooling > Visual Studio 2003)

file-header (executable > 32-bit)

optional-header (subsystem > GUI)

directories (count > 5)

sections (count > 4)

libraries (group > network)

imports (flag > 295)

exports (n/a)

thread-local-storage (n/a)

.NET (n/a)

resources (language > flag)

strings (count > 13063)

debug (stamp > Apr.2008)

manifest (n/a)

version (language > flag)

certificate (n/a)

overlay (n/a)

library (7)	duplicate (0)	flag (1)	first-thunk-original (INT)	first-thunk (IAT)	type (1)	imports (295)	group
msvcrt.dll	-	-	0x000308A8	0x000014A4	implicit	4	-
KERNEL32.dll	-	-	0x0003054C	0x00001148	implicit	48	-
ADVAPI32.dll	-	-	0x00030404	0x00001000	implicit	8	-
USER32.dll	-	-	0x0003064C	0x00001248	implicit	150	-
GDI32.dll	-	-	0x00030428	0x00001024	implicit	72	-
ATMLIB.dll	-	-	0x00030610	0x0000120C	implicit	3	-
DNSAPI.dll	-	×	0x00030620	0x0000121C	implicit	10	network

Libraries loaded msvcrt.dll - Standard C runtime

1. KERNEL32.dll - Process/memory/IO handling
2. ADVAPI32.dll - Registry and privilege management
3. USER32.dll - GUI, input handling
4. GDI32.dll - Graphics manipulation
5. ATL.dll - Active Template Library (COM-related)
6. DNSAPI.dll - Domain Name System networking

Imports count: 295 functions

Functional significance:

- **Process and memory control (KERNEL32.dll):**
 - Likely includes CreateRemoteThread, VirtualAllocEx, WriteProcessMemory → process injection
- **Privilege/registry manipulation (ADVAPI32.dll):**
 - RegCreateKeyEx, RegSetValueEx → persistence in registry
- **Networking (DNSAPI.dll):**
 - DnsQuery_A, DnsQuery_W → DNS lookups for C2 servers
- **GUI manipulation (USER32.dll):**
 - SetWindowsHookEx, MessageBoxA → interaction or injection via hooks

Flagged APIs

Analyzing the flagged API - there are list of APIs

API	Function	Significance
CreateProcess	Execution	Launches external programs
QueueUserWorkItem	Process Injection	Injects code via threads
MiniDumpWriteDump	Memory Dumping	Dumps process memory (e.g., for credential theft)
Sleep	Sandbox Evasion	Delays execution to evade analysis
CreateFile, ReadFile	File Operations	Creates/reads files (payload or config)
GetSystemTimeAsFileTime	Environment Check	Detects VMs or sandboxes via time checks

Step 4: File Hashing

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

FLARE-VM 22-09-2025 16:10:08.58
C:\Users\hrp\Downloads>file xpajB.exe
xpajB.exe: PE32 executable (GUI) Intel 80386, for MS Windows

FLARE-VM 22-09-2025 16:10:15.00
C:\Users\hrp\Downloads>cerutil -hashfile xpajB.exe
'cerutil' is not recognized as an internal or external command,
operable program or batch file.

FLARE-VM 22-09-2025 16:10:35.91
C:\Users\hrp\Downloads>certutil -hashfile xpajB.exe
SHA1 hash of xpajB.exe:
a2e241e9af346714e93c0600f160d05c95839768
CertUtil: -hashfile command completed successfully.

FLARE-VM 22-09-2025 16:10:48.38
C:\Users\hrp\Downloads>_
```

The file xpajB.exe was identified as a 32-bit Windows Portable Executable (PE32) through the file utility, indicating it is a GUI application compiled for Intel 80386 architecture. To further analyze the sample, the built-in Windows tool certutil was used with the -hashfile option to

compute its cryptographic hash. The operation successfully generated a SHA-1 hash value of 2e241e9af346714e93c0600f160d05c95839768. This hash can be leveraged to verify file integrity, cross-reference the executable against threat intelligence databases such as VirusTotal, and determine if the file matches known malicious samples.

Hash Values

- **SHA-256:** e04c85cd4bffa1f5465ff62c9baf0b29b7b2faddf7362789013fbac8c90268bf
- **MD5:** bd76fc01deed43cd6e368a1f860d44ed

VirusTotal

These are the results when we put the following hash in the Virustotal website.

The screenshot displays the VirusTotal analysis interface for a file. The top section shows the file's SHA-256 hash and a community score of 64/72. Below this, the file is identified as `nmas.dll` (520.00 KB) and was last analyzed 17 days ago. The analysis indicates that 64 out of 72 security vendors flagged the file as malicious. The file is categorized as `trojan.xpaj/agent2` and is associated with threat categories `trojan` and `virus`. Family labels include `xpaj`, `agent2`, and `goblin`. The security vendors' analysis table shows the following results:

Security Vendor	Detection
AhnLab-V3	Trojan.Win32.Agent2.C935336
AliCloud	Trojan:Win/Goblin.D
Antiy-AVL	Virus:Win32.Xpaj.gen
Arctic Wolf	procmon Unsafe
Alibaba	Virus:Win32/Goblin.3c3ae782
ALYac	Win32.XPaj.D.1
Arcabit	Win32.XPaj.D.1
Avast	Win32:Gobline [Inf]

virustotal.com/gui/file/812af0ec9e1dfd8f48b47fd148baf6e6ecb42d0a304bc0e4539750dd23820a7f

Set Google Chrome as your default browser and pin it to your taskbar **Set as default**

812af0ec9e1dfd8f48b47fd148baf6e6ecb42d0a304bc0e4539750dd23820a7f

60 / 71
Community Score -56

60/71 security vendors flagged this file as malicious

Reanalyze Similar More

812af0ec9e1dfd8f48b47fd148baf6e6ecb42d0a304bc0e45...
SGDownload.exe

Size 532.14 KB
Last Analysis Date 1 month ago

peexe overlay checks-network-adapters long-sleeps direct-cpu-clock-access runtime-modules

EXE

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 14+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label virus.floxf/pioneer Threat categories virus trojan Family labels floxf pioneer fixflo

Security vendors' analysis Do you want to automate checks?

AhnLab-V3	Win32/Fixflo.GEN	Alibaba	Virus:Win32/Floxf.be59bc24
AliCloud	Virus:Win/Floxf.H	ALYac	Win32.Floxf.A
Antiy-AVL	Virus/Win32.Pioneer.bv	Arcabit	Win32.Floxf.A
idat Wolf	Unsafe	Avast	Win32:FloxfLib-A [Trj]

Conclusion

The static analysis of the suspicious PE file using PEStudio indicates that the binary is most likely unpacked and reveals a rich set of imports that point toward malicious capabilities. The flagged APIs highlight functions associated with the execution of external processes, potential process injection, credential dumping, file system operations, and anti-analysis techniques. While no exports are present—consistent with an EXE rather than a DLL—the observed imports suggest that the sample is designed to act as a backdoor or infostealer, capable of staging additional payloads, harvesting sensitive information, and evading detection. Overall, the malware exhibits behaviors commonly associated with credential theft, persistence, and possible command-and-control communication, making it a high-risk threat.